UNIT 2 — EXAM NOTES

Cloud Computing & Virtual Data Storage Networking

1. DEFINITIONS

Cloud Computing

- The delivery of computing services (storage, servers, databases, software) over the internet — "the cloud."
- Users access resources remotely from data centres managed by providers, instead of owning local hardware.
- Key idea: on-demand, internet-based, pay-as-you-go.

Cloud Storage

- Data stored on remote servers accessed via the internet — not on a local hard drive.
- Introduced to the mainstream around 2006; Google Drive launched in 2012.
- Natural evolution of: physical disks → mapped drives → USB drives → cloud.

Virtualization

- Using software to simulate hardware so one physical machine functions as many virtual ones.
- Enables running multiple Virtual Machines (VMs) on a single physical server.

Data Storage Networking

- Infrastructure (hardware, software, protocols) connecting servers and storage so data can be written, read, and shared across networks.

★ **EXAM TIP** Know all four definitions. Exams often ask you to distinguish cloud computing from traditional IT: Cloud = remote + internet + on-demand. Traditional IT = local hardware + manual management.

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2. KEY CONCEPTS

A. The Three Cloud Service Models (IaaS / PaaS / SaaS)

Model	What It Provides	Real Example
IaaS — Infrastructure as a Service	Raw computing resources: servers, storage, networking. User manages OS & apps.	Amazon Web Services (AWS)
PaaS — Platform as a Service	A development environment to build/run apps. Provider manages hardware + OS.	Microsoft Azure, Facebook

SaaS — Software as a Service	Ready-to-use software over the internet. Provider manages everything.	Google Apps, Ex Libris Alma

★ **EXAM TIP** Memory trick — IaaS = Foundation (the building), PaaS = Workshop (the tools), SaaS = Finished Product (you just use it).

B. Cloud Deployment Models

Shared infrastructure, open to all, managed by provider. Cost-effective. Example: Google Drive: Public Cloud

Dedicated to one organisation. More control and security. Example: In-house hospital system: Private Cloud

Mix of public and private. Example: Library of Congress (cloud for working docs, physical for sensitive archives): Hybrid Cloud

C. Types of Storage

Storage Type	How It Works	Example
DAS — Direct Attached Storage	Connected directly to one computer. Not shared across a network.	External hard drive on a laptop
NAS — Network Attached Storage	Shared across a local network via file-sharing protocols.	Office shared file server
SAN — Storage Area Network	High-speed dedicated network treating all storage as one pool.	Enterprise database storage
Cloud Storage	Accessed over the internet from remote data centres worldwide.	Google Drive, OneDrive, Dropbox

★ **EXAM TIP** DAS = one machine only. NAS = local network. SAN = high-speed network pool. Cloud = internet-based. These distinctions are frequently tested.

D. Types of Virtualization

One physical server runs multiple VMs, each acting as an independent server: Server Virtualization

Multiple physical storage devices managed as one unified pool: Storage Virtualization

Network resources abstracted and divided as needed: Network Virtualization

Desktop environments hosted on remote servers; users access them over a network: VDI (Virtual Desktop Infrastructure)

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3. DETAILED EXPLANATION

How Cloud Storage Works

- You upload a file → it travels over the internet to a data centre → saved on remote servers.
- When needed → you request it → it is sent back over the internet.
- Data centres can be in another city or country entirely.

- Cloud providers use SSL/TLS encryption and two-factor authentication to protect data.

Cloud Service Characteristics (NIST Framework)

Users access resources on demand with no human intervention from the provider: Self-service
Available from any device with an internet connection: Broad Network Access
Provider serves multiple users from shared infrastructure: Resource Pooling
Resources scale up or down automatically based on demand: Elasticity
Usage is monitored and can be reported or billed precisely: Measurability

The Information Factory Concept

- IT systems should work like a factory: reliable, scalable, efficient, and cost-effective.
- The "product" is information services; the "machinery" is servers, storage, and networking.
- Can be public, private, or hybrid — like how a factory can be owned, shared, or outsourced.
- Key attributes: multitenancy, accountability, scalability, rapid provisioning, security.

Cloud Tools in Libraries — How They Fit

Library Function	Cloud Tools Used
Data Storage	Microsoft OneDrive, Google Drive, Dropbox
Documentation & Co-authoring	Google Docs, Microsoft 365, Dropbox Paper
Data Management	Airtable, Google Sheets, Google Forms
Data Visualization	Tableau, Google Sheets, LibInsight
Project Management	Trello, Microsoft Teams, Asana
Communication	Slack, Zoom, Microsoft Teams
Library Management System	Ex Libris Alma, Koha, FOLIO

4. EXAMPLES

Service Model Examples

Netflix uses AWS to run its entire video streaming infrastructure: IaaS — AWS:
Companies build and deploy custom apps on Azure without managing the underlying servers: PaaS — Microsoft Azure:
Staff open a browser and use Docs/Sheets/Drive — no installation, no local maintenance: SaaS — Google Apps:
Libraries access a cloud-hosted library services platform; the vendor manages all the infrastructure: SaaS — Ex Libris Alma:

Cloud Storage Tool Comparison

Tool	Free Storage	Best For
Microsoft OneDrive	5 GB personal / 1 TB Business	Microsoft Office integration, collaborative editing
Google Drive	15 GB free	Docs/Sheets/Forms; easy link-sharing
Dropbox	2 GB free	Simple file sync and sharing across devices

Real-World Scenario Examples

Hybrid cloud — working documents on cloud; sensitive archival records on physical/local storage: Library of Congress

Use OneDrive/Google Drive to access and co-author collection policies from home — no VPN needed: Remote library staff

Files belong to the team, not the individual — when staff leave, documents stay accessible: Google Shared Drives

Uses IaaS (AWS) to store product databases, process transactions, and scale during peak seasons: Amazon E-commerce

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5. ADVANTAGES

Advantages of Cloud Computing

Access files/apps from any device, anywhere, via the internet: Accessibility

Scale storage and computing up or down — only pay for what you use: Scalability

No expensive hardware to buy or maintain; freemium/subscription pricing: Cost Savings

Multiple users edit the same file simultaneously in real time: Collaboration

Data backed up in multiple locations; survives local hardware failure or natural disaster: Disaster Recovery

Provider manages software updates and security patches: Automatic Updates

Seamless access for hybrid and remote workers — no office network required: Remote Work Support

Outsourcing infrastructure frees up internal IT staff: Reduced IT Complexity

Advantages of Virtualization

One physical server runs many VMs — fewer machines needed: Resource Efficiency

VMs can be created, cloned, or moved in minutes: Agility

A crash in one VM does not affect other VMs on the same machine: Isolation

Manage logical (virtual) units rather than individual physical devices: Simplified Management

VMs can be migrated or replicated for disaster recovery: Business Continuance

Disadvantages (Also Examinable)

Data online is vulnerable to hacking, credential theft, and data breaches: Security Risks

If internet or provider is down, you cannot access data: Downtime Dependency

Subscription fees accumulate; scaling adds cost: Ongoing Costs

Requires a reliable, high-speed internet connection: Bandwidth Requirement

After a breach, full recovery can take a very long time: Data Recovery Time

Switching providers is difficult and potentially expensive: Vendor Lock-in

★ **EXAM TIP** Exams often ask for THREE advantages and THREE disadvantages. Learn both sides — a one-sided answer usually scores fewer marks.

6. APPLICATIONS

Library & Information Services

- Collection management workflows: acquisitions, cataloguing, e-resource management, collection assessment.
- Cloud tools streamline hybrid/remote work for librarians.
- Google Forms collects patron feedback and tracks resource trial requests.
- Cloud-based Library Management Systems (LMS): Ex Libris Alma, Koha, FOLIO — browser-based, no local install.
- 1 TB of OneDrive storage per employee ≈ 6.5 million Office/PDF documents.

General Business & IT

Organisations replicate data to cloud for off-site backup: Backup & Disaster Recovery
Businesses host on AWS or Azure instead of buying servers: Website & App Hosting
Online retailers manage product catalogues and transactions in the cloud: E-Commerce
Patient records stored in secure, compliant cloud environments: Healthcare
Cloud-based LMS platforms (Canvas, Moodle) used by universities: Education
Banks store transaction records and run analytics in the cloud: Financial Services
Microsoft 365 and Google Workspace provide cloud email and productivity: Email & Collaboration

When to Use Each Cloud Deployment Model

Scenario	Best Deployment Model
Small business, cost-conscious, non-sensitive data	Public Cloud
Government, hospital, military — sensitive/regulated data	Private Cloud
Large enterprise needing flexibility + control	Hybrid Cloud
Library storing both working docs and rare archival records	Hybrid Cloud

7. SUMMARY & QUICK REVISION

At-a-Glance: Key Points

- Cloud computing = using the internet to access computing resources stored on remote servers.
- Three service models: IaaS (infrastructure) → PaaS (platform) → SaaS (software).
- Three deployment models: Public (shared) | Private (dedicated) | Hybrid (mix).

- Four storage types: DAS (direct) | NAS (local network) | SAN (high-speed pool) | Cloud (internet).
- Key cloud characteristics: self-service, elasticity, multitenancy, measurability, broad access.
- Main advantages: accessibility, scalability, cost savings, collaboration, disaster recovery.
- Main disadvantages: security risks, downtime, ongoing costs, bandwidth needs, vendor lock-in.

Glossary of Key Terms

TERM	DEFINITION
Cloud Computing	Using the internet to access computing resources (storage, apps, servers) on remote servers.
Cloud Storage	Storing data on remote servers accessed via the internet instead of locally.
Virtualization	Using software to simulate hardware; running multiple VMs on one physical machine.
IaaS	Infrastructure as a Service — renting raw computing infrastructure (e.g. AWS).
PaaS	Platform as a Service — renting a development/deployment platform (e.g. Azure).
SaaS	Software as a Service — using fully managed software over the internet (e.g. Google Apps).
Scalability	Ability to increase or decrease resources as demand changes.
Elasticity	Automatic scaling of resources to match workload.
Multitenancy	Multiple users sharing the same underlying infrastructure securely.
DAS	Direct Attached Storage — storage connected directly to one machine only.
NAS	Network Attached Storage — shared storage across a local network.
SAN	Storage Area Network — high-speed dedicated network of storage treated as one pool.
Hybrid Cloud	Combination of public and private cloud environments.
Freemium	Pricing model: basic features free, advanced features paid.
Data Centre	A large facility housing servers and storage hardware for cloud providers.
Vendor Lock-in	Difficulty in switching cloud providers due to dependence on proprietary systems.

★ **EXAM TIP** The three most commonly tested areas are: (1) IaaS vs PaaS vs SaaS — know an example for each. (2) Advantages vs Disadvantages — know at least 3 of each. (3) DAS vs NAS vs SAN vs Cloud — know what distinguishes them.

★ QUICK REVISION SUMMARY

Read this the night before your exam — covers everything essential from Unit 2.

What Is Cloud Computing?

- Accessing computing resources (storage, servers, software) over the internet from remote data centres.
- No need to own or maintain local hardware — pay for what you use, on demand.

- Evolution: physical disks → mapped drives → USB drives → cloud storage (mainstream from 2006).

The 3 Service Models

- **IaaS —** Infrastructure as a Service. Provides raw servers, storage, networking. User manages OS and apps. Example: AWS.
- **PaaS —** Platform as a Service. Development environment to build/run apps. Provider manages hardware + OS. Example: Microsoft Azure.
- **SaaS —** Software as a Service. Ready-to-use software over the internet. Provider manages everything. Example: Google Apps, Ex Libris Alma.

Memory trick: IaaS = the building | PaaS = the workshop | SaaS = the finished product.

Deployment Models

- **Public —** Shared infrastructure, open to all, cost-effective. Best for non-sensitive data.
- **Private —** Dedicated to one organisation. More control and security. Best for sensitive/regulated data.
- **Hybrid —** Mix of both. Best for large organisations needing flexibility and control (e.g. Library of Congress).

Storage Types

- **DAS —** Direct Attached Storage. Connected to one machine only. Not shared.
- **NAS —** Network Attached Storage. Shared across a local network.
- **SAN —** Storage Area Network. High-speed dedicated network; all storage treated as one pool.
- **Cloud —** Accessed over the internet from remote data centres. Examples: Google Drive, OneDrive, Dropbox.

Key Advantages vs Disadvantages

Advantages:

- Accessibility — access from any device, anywhere via the internet.
- Scalability — scale resources up or down; only pay for what you use.
- Cost Savings — no hardware to buy or maintain; freemium/subscription pricing.
- Collaboration — multiple users edit the same file simultaneously in real time.
- Disaster Recovery — data backed up across multiple locations; survives local hardware failure.

Disadvantages:

- Security Risks — vulnerable to hacking, credential theft, and data breaches.
- Downtime Dependency — if internet or provider is down, you cannot access data.
- Ongoing Costs — subscription fees add up; scaling increases cost.
- Bandwidth Needed — requires a reliable, high-speed internet connection at all times.
- Vendor Lock-in — switching providers is difficult and costly once embedded.

Must-Know Facts for the Exam

- Cloud storage was introduced to the mainstream in 2006; Google Drive launched in 2012.
- 1 TB of OneDrive storage per employee ≈ 6.5 million Office/PDF documents.
- Library of Congress uses a hybrid approach — cloud for working docs, physical storage for sensitive archives.
- Google Shared Drives: files belong to the team, not the individual — prevents loss of data when staff leave.
- Cloud costs are typically subscription-based (freemium model: basic free, advanced features paid).
- Virtualization enables agility (not just consolidation) — VMs can be created, moved, or cloned rapidly.

★ **FINAL EXAM TIP** Focus on three areas: (1) IaaS vs PaaS vs SaaS — name and example for each. (2) At least 3 advantages AND 3 disadvantages of cloud computing. (3) What distinguishes DAS, NAS, SAN, and Cloud storage from each other. These come up most often.

Sources: Collection Management in the Cloud (Kipps & Jones, 2022) | Cloud and Virtual Data Storage Networking (Schulz, 2012)